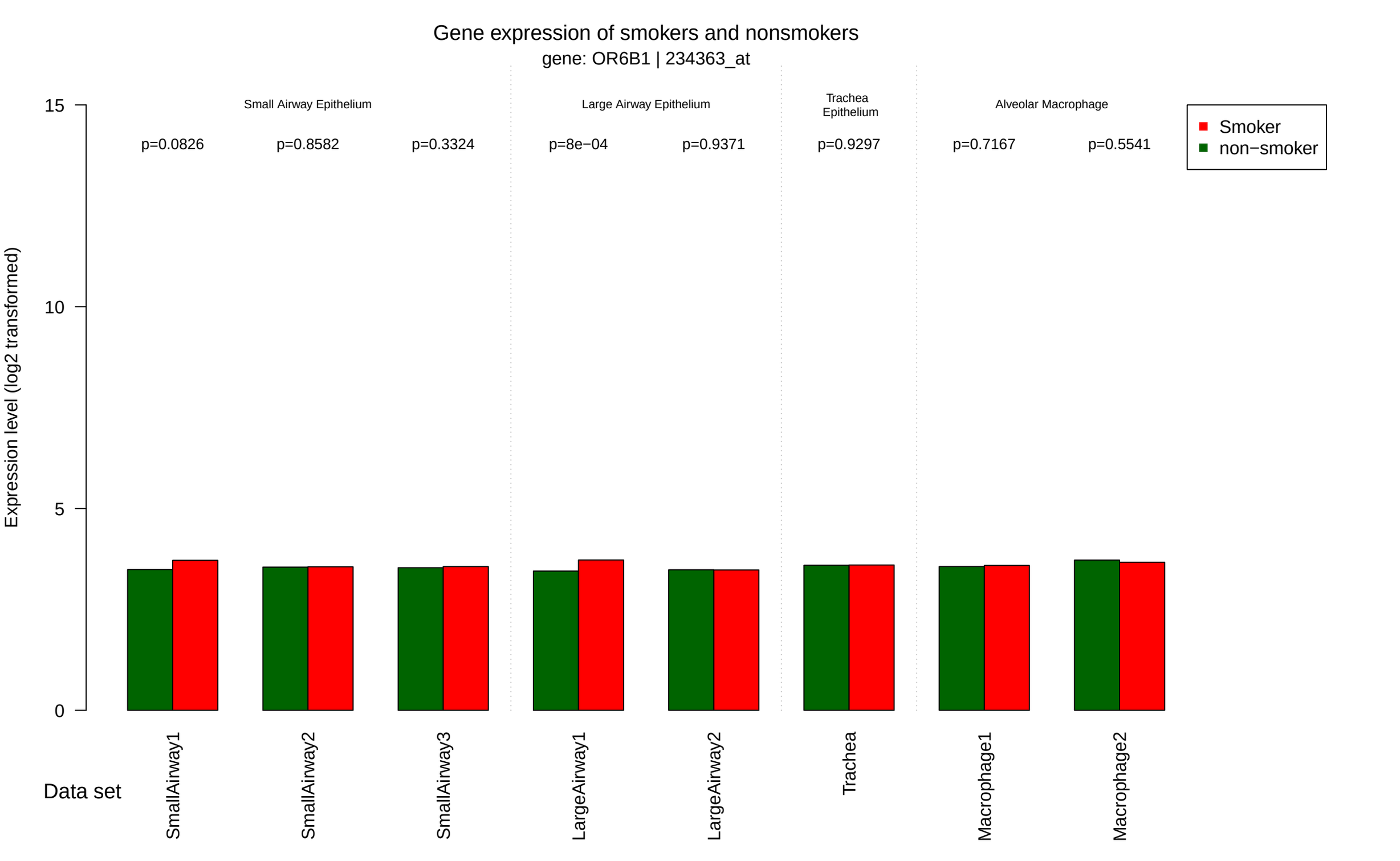
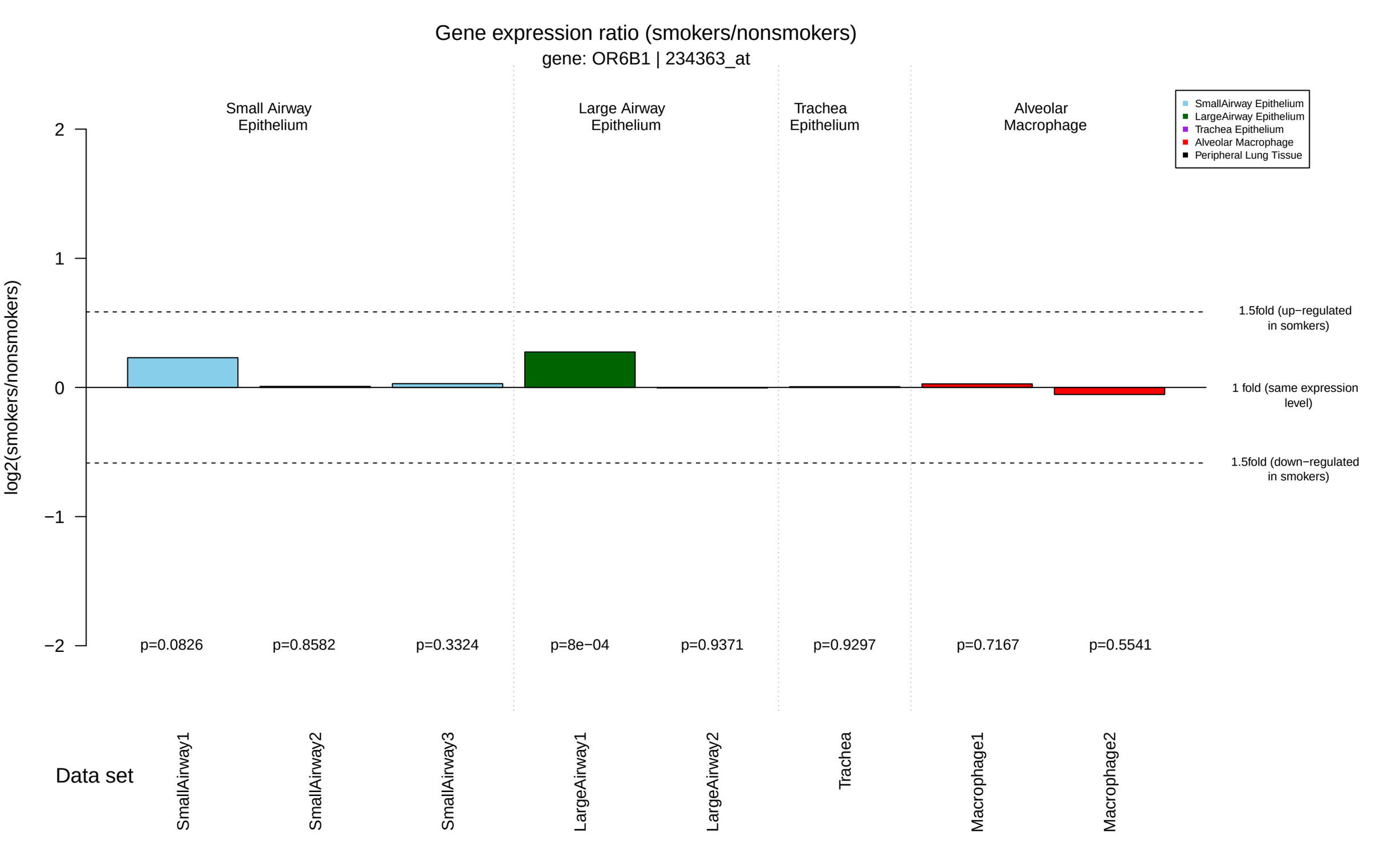
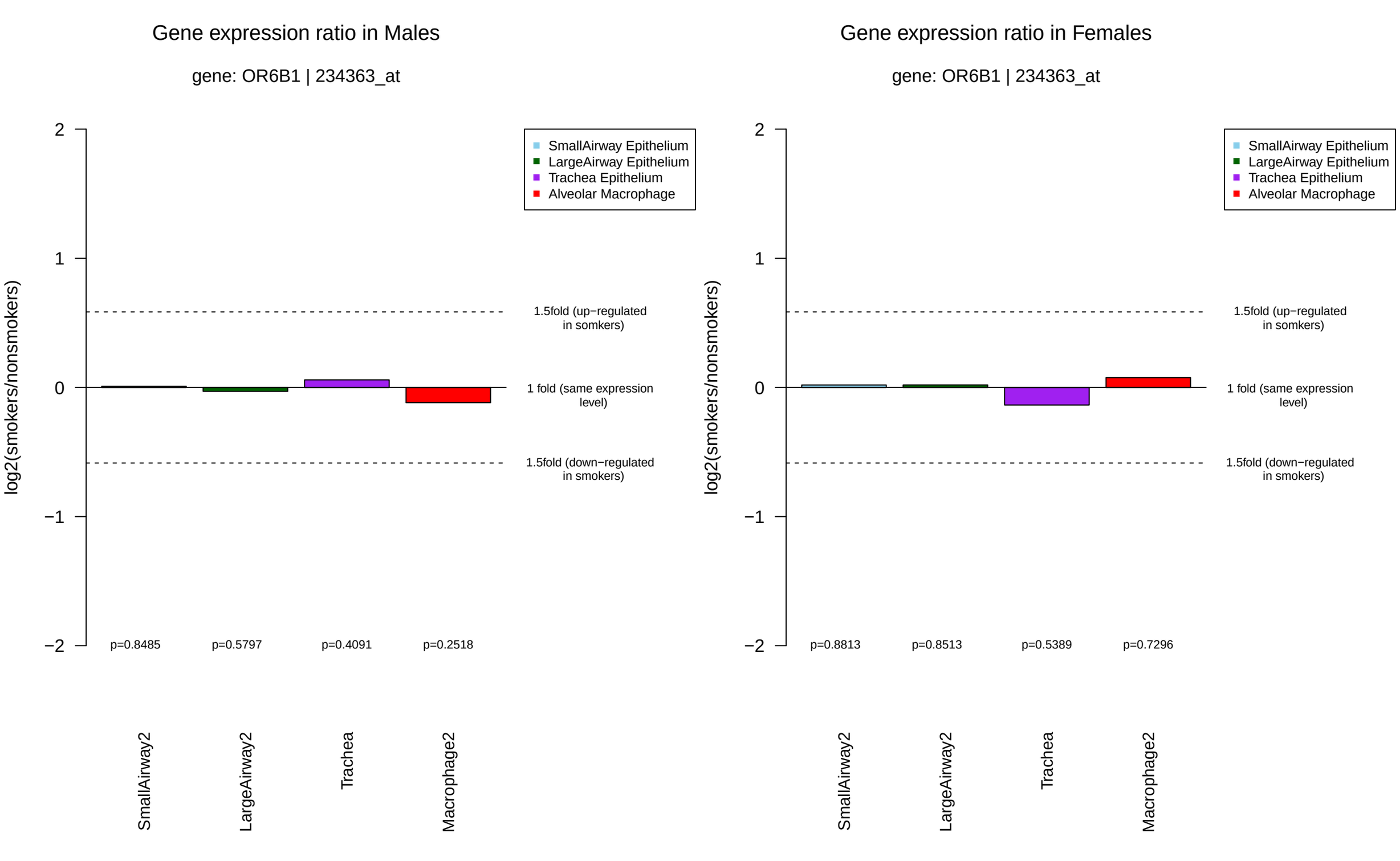
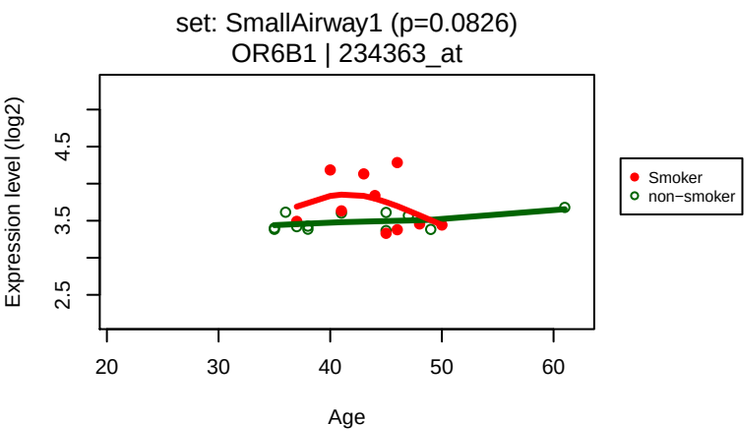
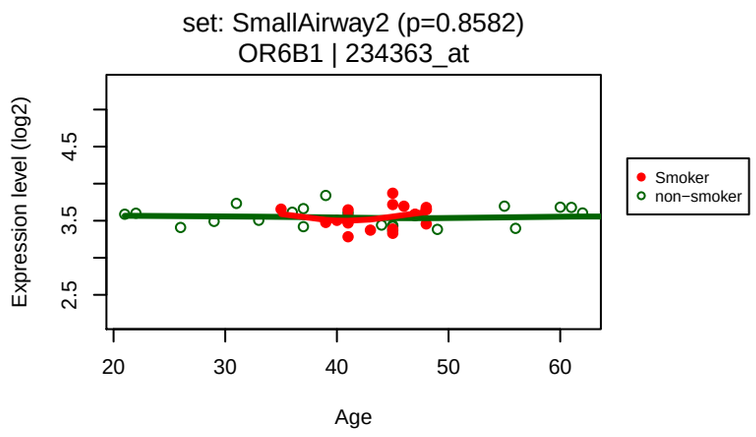
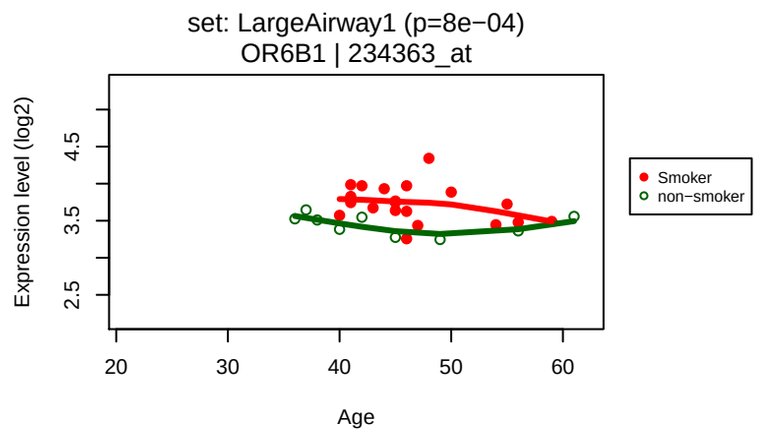

Gene expression of smokers and nonsmokers
gene: OR6B1 | 234363_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0826
p=0.8582
p=0.3324
p=8e−04
p=0.9371
p=0.9297
p=0.7167
p=0.5541
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

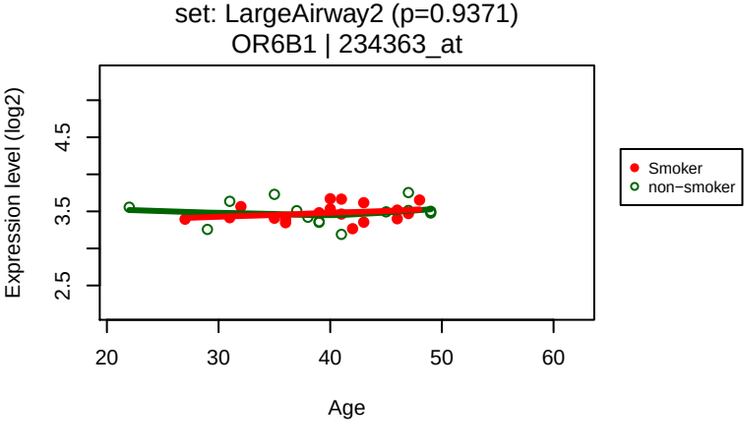

Gene expression ratio (smokers/nonsmokers)
gene: OR6B1 | 234363_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0826
p=0.8582
p=0.3324
p=8e−04
p=0.9371
p=0.9297
p=0.7167
p=0.5541
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

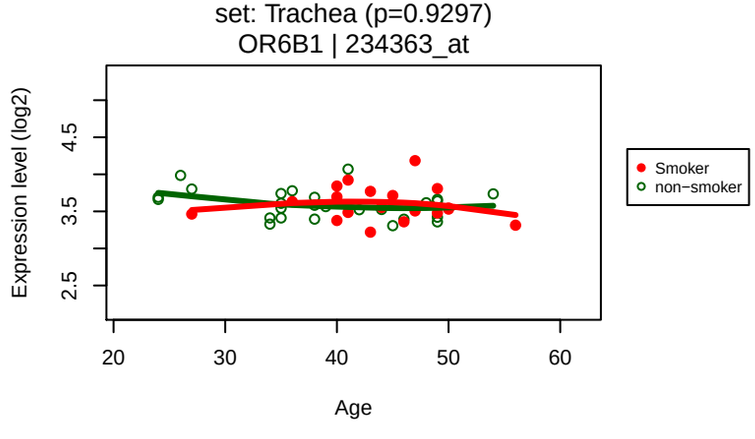

Gene expression ratio in Males
gene: OR6B1 | 234363_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8485
p=0.5797
p=0.4091
p=0.2518
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR6B1 | 234363_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8813
p=0.8513
p=0.5389
p=0.7296
SmallAirway2
LargeAirway2
Trachea
Macrophage2

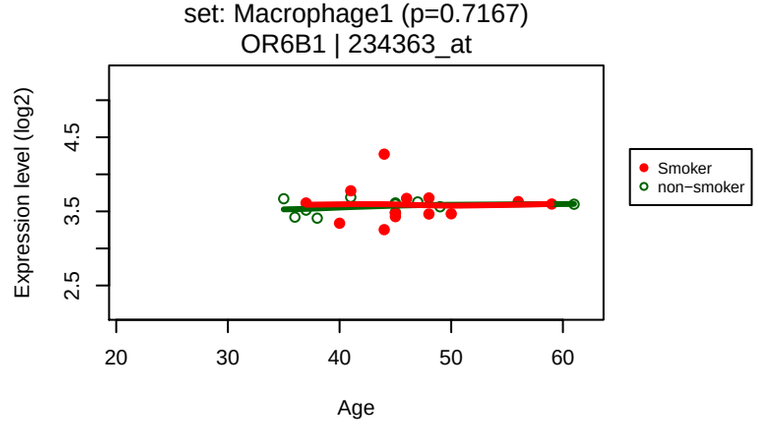

set: SmallAirway1 (p=0.0826)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

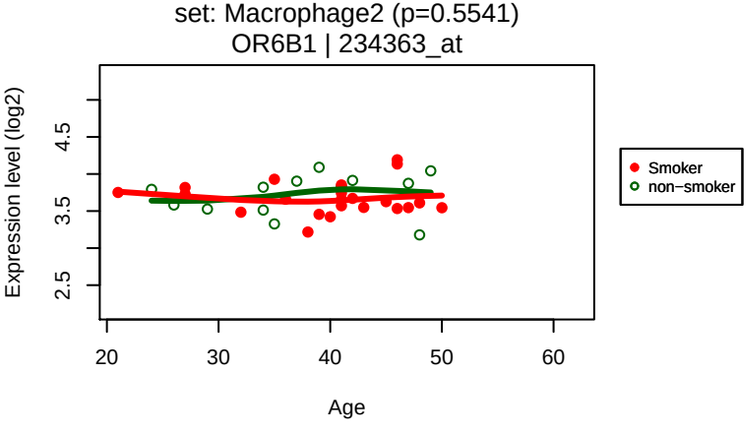

set: SmallAirway2 (p=0.8582)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

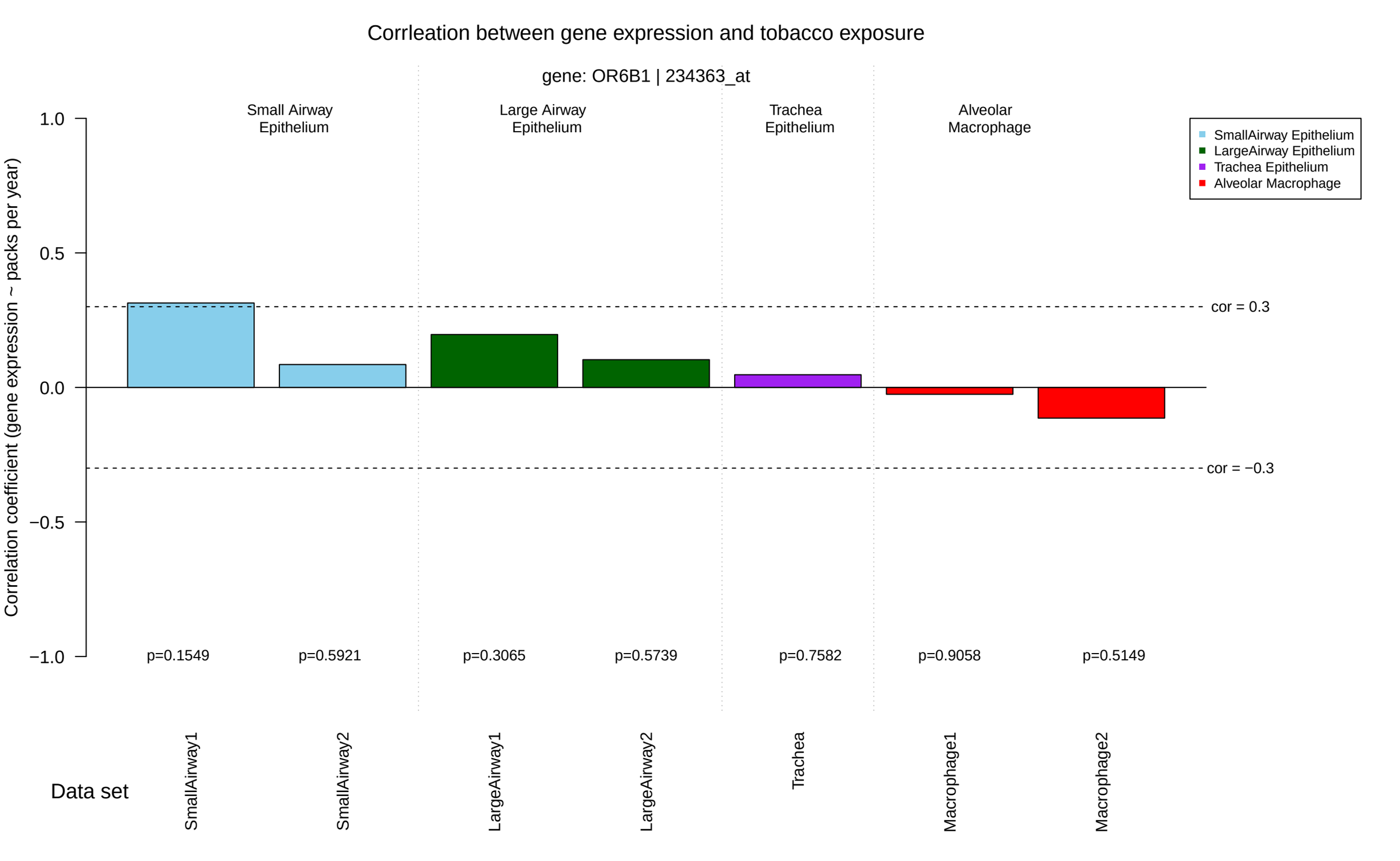

set: LargeAirway1 (p=8e−04)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9371)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.9297)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7167)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5541)
OR6B1 | 234363_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR6B1 | 234363_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1549
p=0.5921
p=0.3065
p=0.5739
p=0.7582
p=0.9058
p=0.5149
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2